12/10/2026

Week - 4

Arrays :- (Homogeneous)

1) The array is a collection of the same type of elements at contiguous memory location under the same name.
2) It has a finite data.
3) It is a easier to access the element in the case of an array
4) Insertion and deletion operations are costly in the case of an array.
5) No modification is possible at the runtime after the array is created.
6) Arrays are static.

List :

1) List are dynamic and can change in size during runtime.

2) List can accommodate elements of different data type. (Non-Homogeneous)

★ Not :- If index sequence is not proper like from 0 to 1, 2, 3, 4... it will give you a index error

eg :- labels = ["R-001", "R-002", "R-003"]
0 1 2
-3 -2 -1

Slicing :- same rule as Range (start, stop)
(stop) is never included.

example :- values = [10, 20, 30, 40, 50]
0 1 2 3 4

print (values [1:3])

output ⟹ (20, 30)

Append = you can add any element at the last of the list.

eg :- x = [1, 2, 3] → print (x)
x.append(20) output ⟹ [1, 2, 3, 20]

Tuple :- A tuple in python is a collection of items that is ordered and cannot be changed after it is created (this is called immutable).

* Note : Tuple = ( ) → round bracket
list = [ ] → square bracket

* Note = list can be change able (mutuable)
Tuple are not (immutable)

Note :- Dictionaries { } → curly brackets.

It stores { — : — , — : — }

key value

pairs

⇒ A dictionary is a data structure that stores information in key - value pairs

eg :- record = { "label" : "Temperature" }

print (record ["label"])

output ⇒ Temperature.

Set :- A set is a collection of unique values. It automatically removes duplicate values and does not maintain a guaranted order. It is mutable.

↓

remove or add element after creating.

Note :- String " " ⟶ Immutable.